EXPERIMENT:4

Apache Hive & Pig for Querying Large Datasets

AIM

To install **Apache Hive** and **Apache Pig**, create tables, load large datasets into HDFS, and execute queries using HiveQL and Pig Latin.

PART 1: PREREQUISITES

Before installing Hive and Pig, ensure the following are installed:

1. Java (JDK 8 or 11)
2. Hadoop (running in pseudo-distributed mode)

You must have:

- **Apache Hadoop** installed
- HDFS working
- NameNode and DataNode running

Check Hadoop:

hdfs dfs -ls /

If it works, proceed.

PART 2: INSTALLATION OF APACHE HIVE (STEP-BY-STEP)

Step 1: Download Hive

Download Hive binary from official website.

Example:

- hive-3.1.3-bin.tar.gz

Step 2: Extract Hive

tar -xvf hive-3.1.3-bin.tar.gz
sudo mv hive-3.1.3-bin /usr/local/hive

Step 3: Set Environment Variables

Edit .bashrc:

```
export HIVE_HOME=/usr/local/hive
export PATH=$PATH:$HIVE_HOME/bin
```

Reload:

```
source ~/.bashrc
```

Step 4: Configure Hive

Go to:

```
cd $HIVE_HOME/conf
```

Copy template:

```
cp hive-default.xml.template hive-site.xml
```

Step 5: Configure Metastore (MySQL or Derby)

For lab (simple setup), use Derby (default).

Initialize schema:

```
schematool -dbType derby -initSchema
```

Step 6: Create HDFS Directory for Hive

```
hdfs dfs -mkdir /user/hive
hdfs dfs -mkdir /user/hive/warehouse
hdfs dfs -chmod g+w /user/hive/warehouse
```

Step 7: Start Hive

```
hive
```

If Hive prompt appears → Installation successful.

PART 3: INSTALLATION OF APACHE PIG (STEP-BY-STEP)

Step 1: Download Pig

Download Pig binary (example: pig-0.17.0.tar.gz)

Step 2: Extract Pig

```
tar -xvf pig-0.17.0.tar.gz
sudo mv pig-0.17.0 /usr/local/pig
```

Step 3: Set Environment Variables

Add to .bashrc:

```
export PIG_HOME=/usr/local/pig
export PATH=$PATH:$PIG_HOME/bin
```

Reload:

```
source ~/.bashrc
```

Step 4: Start Pig

```
pig
```

Pig Grunt shell should appear.

Installation complete.

PART 4: HIVE – CREATION OF TABLES, DATA LOADING & QUERIES

Dataset Example (employees.csv)

```
1,John,50000,IT
2,Alice,60000,HR
3,Bob,45000,Finance
```

Step 1: Upload Dataset to HDFS

```
hdfs dfs -mkdir /input
hdfs dfs -put employees.csv /input/
```

Step 2: Create Hive Table

Start Hive:

```
hive
```

Create table:

```
CREATE TABLE employees (
   id INT,
   name STRING,
   salary INT,
   department STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

Step 3: Load Data into Table

```
LOAD DATA INPATH '/input/employees.csv'
INTO TABLE employees;
```

Step 4: Run Queries

View Data

```
SELECT * FROM employees;
```

Filter Query

```
SELECT name, salary
FROM employees
WHERE salary > 50000;
```

Aggregation Query

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department;
```

SAMPLE OUTPUT (Hive)

```
Alice 60000

IT      50000
HR      60000
Finance 45000
```

PART 5: PIG – DATA LOADING & QUERIES

Step 1: Start Pig

```
pig
```

Step 2: Load Data

```
employees = LOAD '/input/employees.csv'
USING PigStorage(',')
AS (id:int, name:chararray, salary:int, department:chararray);
```

Step 3: Display Data

```
DUMP employees;
```

Step 4: Filter Records

```
high_salary = FILTER employees BY salary > 50000;
DUMP high_salary;
```

Step 5: Group & Aggregate

```
grouped = GROUP employees BY department;
avg_salary = FOREACH grouped GENERATE group, AVG(employees.salary);
DUMP avg_salary;
```

SAMPLE OUTPUT (Pig)

```
(HR,60000.0)
(IT,50000.0)
(Finance,45000.0)
```

RESULT

Apache Hive and Apache Pig were successfully installed and configured.Tables were created in Hive.Data was loaded from HDFS.Queries were executed using HiveQL.Data transformations and aggregations were performed using Pig Latin.Hence, querying large datasets using Hive and Pig was successfully performed.